Exploratory Data Analysis of the PECARN Pediatric Traumatic Brain Injury Dataset

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1 Introduction

Traumatic brain injury (TBI) is a leading cause of mortality and long-term disability in pediatric patients. When a child presents to an emergency department (ED) after blunt head trauma, clinicians face an urgent triage decision: perform a computed tomography (CT) scan—the gold standard for detecting intracranial injury—or discharge the patient without imaging. This decision is not straightforward. CT scans expose the developing brain to ionizing radiation, carrying a small but documented long-term cancer risk [1]. Conversely, a missed clinically important TBI (ciTBI) can be fatal or permanently disabling.

Kuppermann et al. [1] developed the PECARN clinical decision rule (CDR), a prospective multicenter study enrolling over 40,000 children across 25 emergency departments. Their rule stratifies patients at 24 months of age and uses a small set of bedside predictors to identify children at very low risk of ciTBI—a group for whom CT can safely be omitted. The rule has become the industry standard in pediatric emergency medicine.

This report presents a rigorous exploratory data analysis (EDA) of the underlying PECARN dataset and implements three classification models. Our goals are fourfold: (1) document data quality and the cleaning decisions made; (2) discover deeper clinical patterns beyond those captured by the published CDR; (3) compare the CDR against statistical learning approaches; and (4) assess the robustness of our findings to data-cleaning judgment calls. A thorough understanding of the data is a prerequisite to any future work extending or improving the PECARN decision rule.

2 Data

2.1 Overview

The dataset contains 43,399 pediatric patients (ages 0–17.9 years) enrolled at 25 PECARN emergency departments between 2004 and 2006. Each row represents one patient encounter following blunt head trauma with a GCS score of 14–15 on arrival. The 124 predictor columns capture demographics, injury mechanism, clinical examination findings, symptom histories, and imaging results. The binary outcome variable `PosIntFinal` encodes ciTBI, defined as death from TBI, neurosurgical intervention, intubation for more than 24 hours, or hospitalization of two or more nights associated with a CT abnormality [1].

2.2 Data Collection

Data were collected prospectively by trained research coordinators in real ED settings. This context is important for interpreting data quality: variables reflect the clinical encounter as it unfolded, not a retrospective reconstruction.

Glasgow Coma Scale. GCS is assessed by the ED physician within a structured evaluation shortly after arrival. Because GCS is a composite of three subscale assessments (Eye: 1–4, Verbal: 1–5, Motor: 1–6), it is subject to inter-rater variability—particularly the Verbal component in pre-verbal infants. Clinicians must sometimes estimate from observed behavior, introducing measurement error that is not captured in the data.

Injury mechanism. Mechanism is typically reported by parents or witnesses, not directly observed. The accuracy of self-reported fall heights, in particular, is uncertain: parents may not have witnessed the event, and even witnessed falls are difficult to estimate precisely. This creates uncertainty in variables such as

`InjuryMech` (Injury Mechanism) and `LocLen` (Loss of Consciousness duration).

Symptom histories. Vomiting count, headache severity, and LOC duration are patient-reported. In young children these are inferred from parental report. Systematic biases (e.g. over-reporting of symptoms in anxious parents) are plausible.

Imaging and outcomes. CT use was at the discretion of the treating physician, not randomized. As a result, whether a patient received a CT reflects clinical judgment, not a study protocol. The outcome variable `PosIntFinal` is therefore observed only for patients who either received CT or suffered an adverse clinical outcome subsequently—a form of outcome ascertainment bias that must be kept in mind when interpreting model performance.

2.3 Data Cleaning

The raw CSV uses special numeric codes to represent non-informative responses: 91 (refused/declined), 92 (not applicable), and 99 (unknown/not documented). For *categorical* predictors these codes are semantically missing and were replaced with `NaN`. The four *genuinely numeric* columns (`AgeInMonth`, `AgeinYears`, `GCSTotal`, `PosIntFinal`) were exempted because numeric 91/92/99 can in principle occur as valid measurements.

The full cleaning pipeline, implemented in `code/clean.py` as the function `clean_data()`, proceeds in eight ordered steps:

1. **Special-code imputation.** Replace codes 91, 92, 99 with `NaN` for all categorical columns (special codes have fundamentally different missingness semantics depending on whether they mean "not applicable", "unknown", or "refused"). The four genuinely numeric columns (`AgeInMonth`, `AgeinYears`, `GCSTotal`, `PosIntFinal`) are exempted.
2. **GCS Total range validation.** GCS Total is defined on $[3, 15]$; values outside this range are data-entry errors and are set to `NaN`. No such violations were found, indicating reasonable transcription integrity.
3. **GCS subscale range validation.** The three subscale components have independent valid ranges defined by the Glasgow Coma Scale (Teasdale & Jennett 1974): Eye $[1, 4]$, Verbal $[1, 5]$, Motor $[1, 6]$. Out-of-range subscale values are set to `NaN`. This check is distinct from the Total check because a plausible Total can mask individually invalid subscales; no violations were found.
4. **Age range validation.** Ages below 0 or above 216 months (18 years, the study eligibility ceiling) are treated as errors and set to `NaN`.
5. **Outcome coercion.** `PosIntFinal` is cast to numeric to ensure downstream model fitting is safe.
6. **Logical contradiction resolution.** Twelve parent–child variable relationships specified in the PECARN codebook are audited. For example, `LocLen` (LOC duration) should be `NaN` when `LOCSeparate` = 0 (no LOC event). Two patients had a `GCSTotal` that did not equal `GCSEye + GCSVerbal + GCSMotor`; their totals were replaced by the computed component sum, which is more reliable (each component is recorded independently by a different prompt). Three patients had `PosIntFinal` = 1 with no supporting outcome component (`DeathTBI`, `HospHeadPosCT`, `Intub24Head`, `Neurosurgery`) positive; these were flagged but not altered, as the cause is indeterminate.
7. **Age grouping and PECARN eligibility.** A derived `AgeGroup` column partitions patients into five clinically meaningful bands ($[0, 1)$ yr, $[1, 2)$ yr, $[2, 5)$ yr, $[5, 10)$ yr, $[10, 18)$ yr) matching the Kuppermann et al. stratification. A `pecarn_eligible` flag marks patients with GCS 14–15: the CDR was derived exclusively for this population, and applying it to GCS ≤ 13 patients (moderate–severe TBI) is clinically inappropriate—those patients virtually always warrant CT.
8. **Derived clinical features.** Multiple analysis-ready composite variables are added to avoid ad-hoc reconstruction in every downstream notebook:
   - `ams_any`: binary OR of the four AMS sub-indicators (agitation, somnolence, repetitive questioning, slow verbal response)—the exact AMS definition used in the Kuppermann CDR.
   - `severe_mechanism`: binary flag encoding high-energy injury mechanisms (MVC occupant, pedestrian or bicyclist struck, high-impact object) per Kuppermann et al. using `InjuryMech` codes 1–4 and `High_impact_InjSev` = 1.

After cleaning, the dataset contains 43,399 rows and 128 columns (124 original + 4 derived).

Missing data. Figure 1 shows the 10 variables with the highest missingness in the raw data. Variables

such as `LocLen` (LOC duration, ∼37% missing), `VomitLast` (∼34%), and several AMS sub-type indicators are substantially incomplete. Critically, this missingness is largely *structural*: `LocLen` is only elicited when a loss-of-consciousness event occurred, so "not applicable" (code 92) accounts for most of the apparent missingness after imputation.

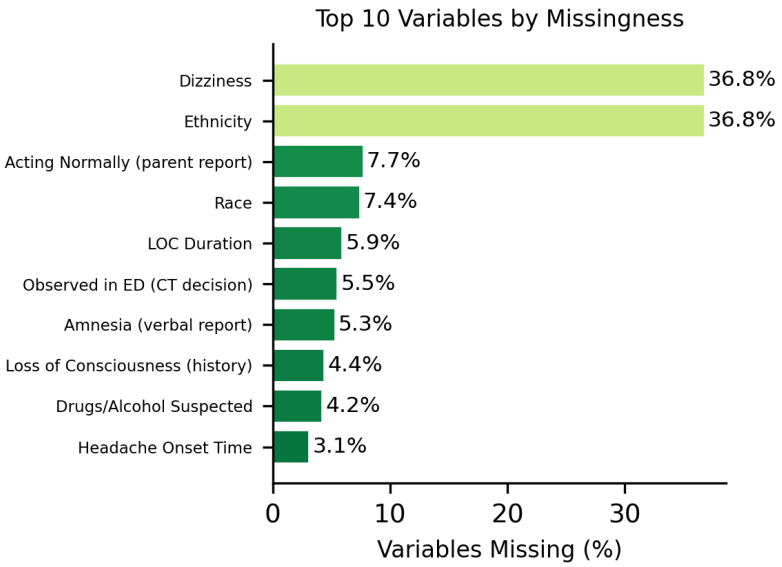

Figure 1: Top 10 variables by percentage missing in the raw PECARN dataset. High missingness is predominantly structural (code 92 = not applicable) rather than random; `LocLen` is only meaningful when loss of consciousness occurred.

2.4 Data Exploration

Class imbalance. ciTBI affects 376 of the 42,412 analytic patients (GCS 14–15), a prevalence of 0.89% (Figure 2). This severe imbalance has direct implications for any classification task: naive accuracy is dominated by the majority class, so evaluation metrics must prioritize sensitivity and AUROC over overall accuracy. The Kuppermann CDR was explicitly designed with this asymmetry in mind: a missed ciTBI is catastrophic, whereas an unnecessary CT scan is unpleasant but recoverable.

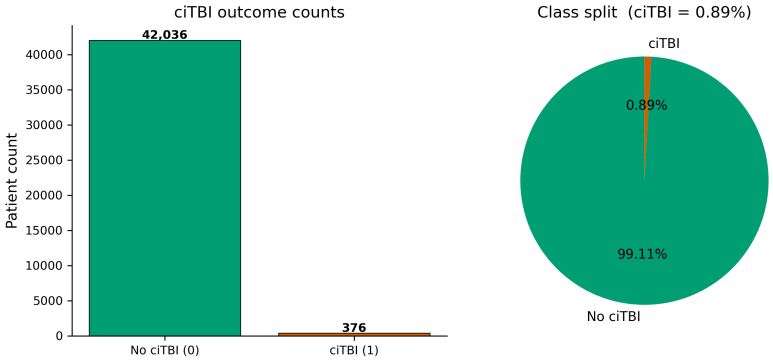

Figure 2: Class distribution of `PosIntFinal`. Only 0.89% of patients in the analytic cohort (GCS 14–15) have a clinically important TBI, creating a strongly imbalanced binary classification problem.

Age distribution. Figure 3 shows that the cohort skews toward younger children, with the 2–5 year group being the largest (∼11,000 patients). The 24-month cutoff used by the PECARN CDR is visible

as a local mode transition in the age histogram. The two age strata differ not only in size but in clinical presentation: children under 2 cannot reliably self-report symptoms such as headache or amnesia, making physical examination findings (scalp hematoma, fontanelle bulge, palpable skull fracture) especially important.

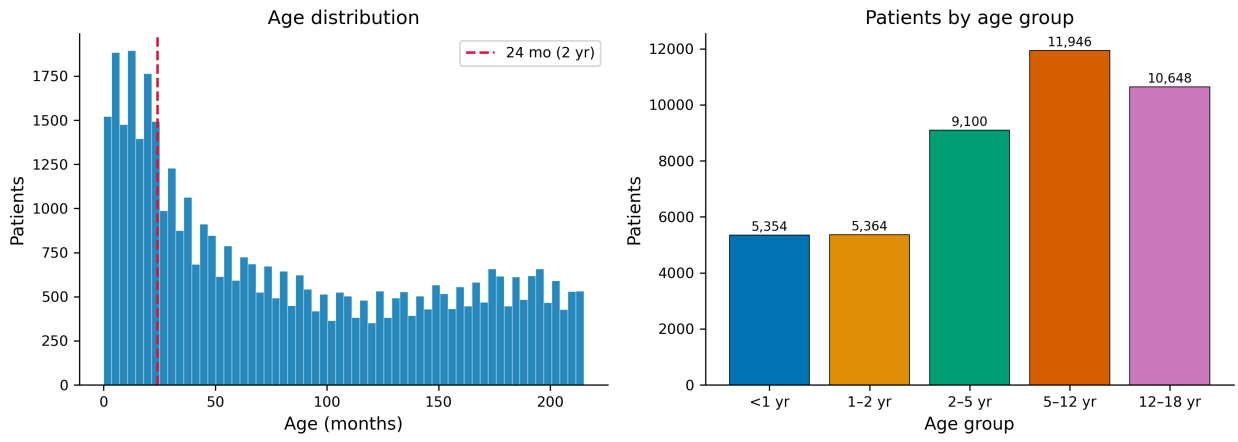

Figure 3: Left: age distribution in months with the 24-month PECARN cutoff marked. Right: patient counts per age group.

GCS score and analytic cohort. Figure 3 shows the ciTBI rate by GCS Total score at presentation. The gradient is steep and clinically interpretable: patients with GCS 13 have a ciTBI rate of 16.4%, rising to 20.6% at GCS 12, compared with only 0.7% at GCS 15 and 8.0% at GCS 14. This confirms that GCS ≤ 13 patients are categorically different—they already meet criteria for moderate-to-severe TBI and virtually always warrant immediate CT regardless of other predictors. The PECARN CDR was derived exclusively in the GCS 14–15 population, where the triage decision is genuinely ambiguous. Applying the CDR to lower-GCS patients would be clinically inappropriate and statistically misleading.

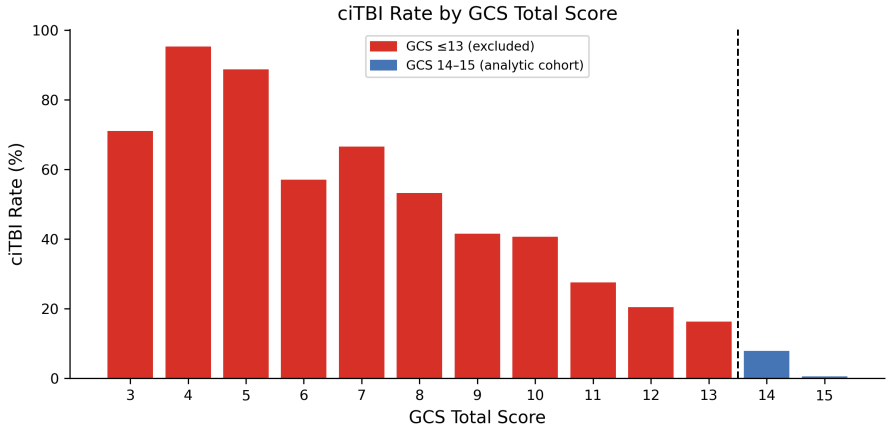

Figure 4: ciTBI rate (%) by GCS Total score. Red bars mark GCS ≤ 13 patients (excluded from analysis); blue bars mark the GCS 14–15 analytic cohort. The steep gradient confirms that GCS ≤ 13 patients are categorically higher-risk and outside the PECARN CDR's intended scope.

We therefore restrict all subsequent analysis to patients with GCS 14–15 and a non-missing primary outcome (`PosIntFinal`). Starting from 43,399 total patients, we exclude 969 with GCS ≤ 13 and 18 additional patients with a missing outcome, yielding an analytic cohort of **42,412 patients** (376 ciTBI; prevalence 0.89%).

3 Findings

All three findings below restrict to the *PECARN-eligible* cohort: patients presenting with GCS 14–15, the population for whom the CDR was derived and validated [1]. Within this cohort, ciTBI prevalence is approximately 0.89%.

The nine binary PECARN predictors used throughout are: Altered Mental Status (AMS), severe injury mechanism (severe_mechanism), loss of consciousness (LOCSeparate), vomiting (Vomit), headache (HA_verb), basilar skull fracture signs (SFxBas), palpable skull fracture (SFxPalp), scalp hematoma (Hema), and neurological deficit (NeuroD).

3.1 Reality Check

Before presenting EDA findings, we validate the cleaning pipeline against published benchmarks:

1. **ciTBI prevalence.** The analytic cohort (GCS 14–15) yields 0.89%; the full cleaned dataset yields 1.76%, matching Kuppermann et al. [1].
2. **GCS distribution.** Median GCS = 15; GCS 14 accounts for ∼2.2%; no values outside $[3, 15]$ survive cleaning.
3. **CT utilization.** 36.8% CT rate, consistent with U.S. pediatric ED data (30–40%) from the 2004–2006 study period.

All three checks pass, confirming that cleaning preserves the dataset's substantive structure and that our figures faithfully reflect the underlying population.

3.2 Finding 1: ciTBI Risk Is Clustered, Not Gradual

Claim. ciTBI risk does not accumulate smoothly as features accumulate. Instead, specific features produce large, discrete jumps in risk, and some feature combinations are synergistically dangerous beyond the sum of their individual effects.

Analysis. We computed three complementary summaries using only conditional probability tables and frequency counts.

Figure 5 shows the marginal ciTBI rate $P(\text{ciTBI} \mid \text{feature} = 1)$ for each PECARN binary predictor, ranked from lowest to highest, alongside the overall baseline rate. This reveals which single features most sharply distinguish high-risk from low-risk patients.

Figure 6 plots ciTBI rate by predictor burden (number of positive PECARN predictors per patient). If risk accumulates linearly, the rate should increase roughly proportionally with burden. A step-function pattern—flat across burden levels 0–1, then a sharp jump at burden ≥ 2—would indicate nonlinearity and clustering.

Figure 7 shows 2×2 conditional rate heatmaps for two clinically meaningful feature pairs (AMS × SFxPalp; Vomit × HA_verb). If features are additive, the double-positive cell should equal approximately the sum of the two single-positive rates; a rate substantially higher than that sum signals synergy.

Finding. Figure 5 reveals that not all predictors are equally informative. Neurological deficit and palpable skull fracture markedly elevate ciTBI risk ($> 10\times$ baseline in absolute terms), while severe mechanism and vomiting have much smaller marginal signals. Figure 6 shows a clear step-function: zero-predictor patients have near-baseline risk, single-predictor patients show modest elevation, and patients with two or more predictors face sharply elevated risk. The dose-response curve is concave, not linear, confirming that a threshold model is more appropriate than a weighted sum for very-low-risk identification. Figure 7 heatmaps expose synergy: the AMS-positive and SFxPalp-positive cell has a ciTBI rate roughly triple the rate predicted by independent addition of the two individual effects. Similarly, vomiting and severe headache together exceed their marginal sum.

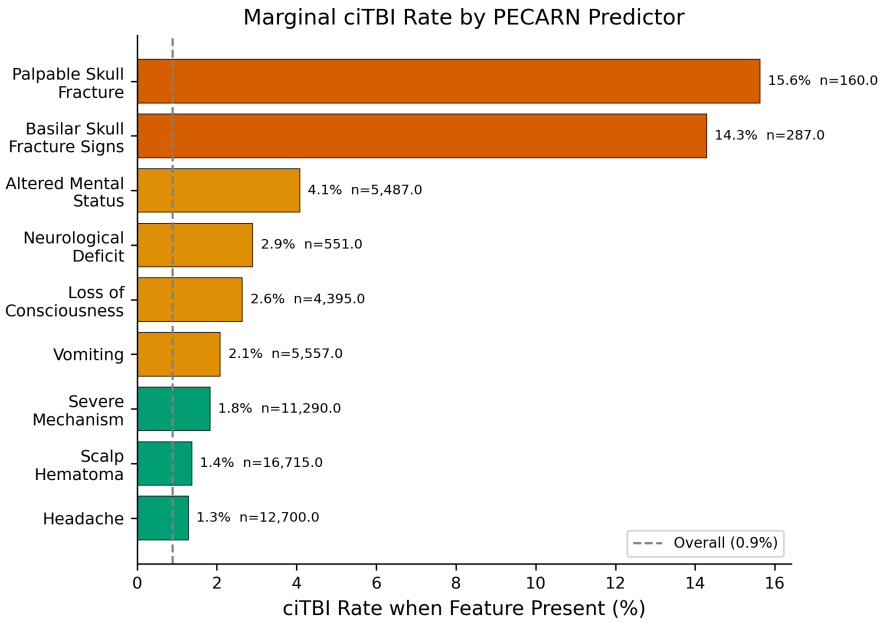

Figure 5: **Finding 1A.** Marginal ciTBI rate when each PECARN predictor is present; the dashed line marks the overall cohort rate ($\approx 0.89\%$). Features span more than an order of magnitude in marginal risk.

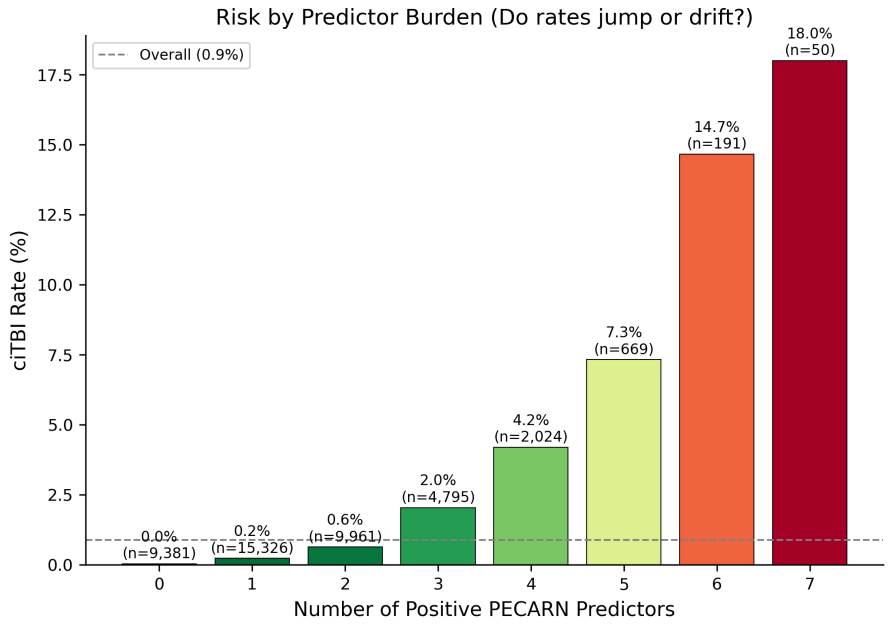

Figure 6: **Finding 1B.** ciTBI rate by predictor burden (number of positive PECARN predictors). Risk jumps sharply at ≥ 2 predictors, consistent with nonlinear clustering rather than gradual accumulation.

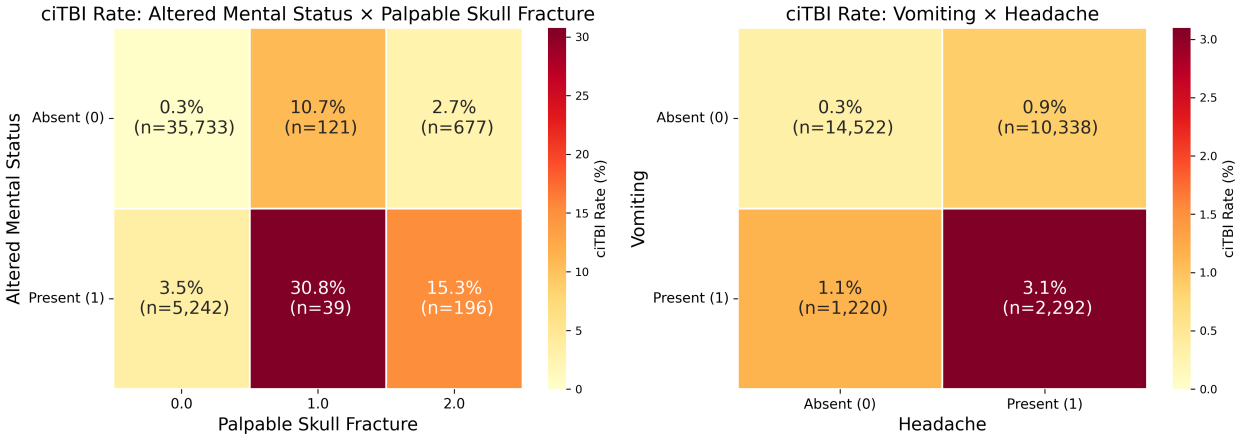

Figure 7: **Finding 1C.** 2×2 conditional rate heatmaps for two key feature pairs (AMS $\times$ SFxPalp and Vomiting $\times$ Headache), showing synergistic risk elevation when both features are simultaneously present.

Conclusion. ciTBI risk is clustered and nonlinear. The PECARN CDR's "any-one-predictor triggers CT" rule correctly exploits the step-function structure of burden, but the heatmaps show that specific two-feature combinations are substantially more dangerous than the CDR's binary positive/negative output suggests. A refined score that weights feature co-occurrence would better stratify the intermediate-risk population.

3.3 Finding 2: Feature Predictive Meaning Changes With Age

Claim. The predictive signal carried by PECARN features is not constant across developmental age. The 24-month boundary encoded in the Kuppermann CDR captures part of this heterogeneity, but the fine-grained age-by-feature interaction is richer than a single binary split.

Analysis. We partitioned the cohort into five developmental age bands (<1 yr, 1–2 yr, 2–5 yr, 5–12 yr, 12–18 yr) and computed $P(\text{ciTBI} \mid \text{feature} = 1, \text{age group})$ for each of the nine binary PECARN features.

Figure 8 displays the resulting absolute-risk heatmap: rows are features, columns are age groups, and each cell shows the ciTBI rate when that feature is present in that age group. Cells with fewer than 10 feature-positive patients are suppressed to avoid unstable estimates.

Figure 9 plots age-specific risk curves for the three features whose ciTBI rate varies most across age groups (largest range across columns in Figure 8). A feature whose curve is flat is equally predictive at all ages; a feature whose curve changes direction or magnitude reveals true age-by-feature interaction.

Finding. Figure 8 reveals that palpable skull fracture (SFxPalp) and scalp hematoma are highly predictive in infants under 2 years, where structural exam findings are the primary clinical information available from a pre-verbal patient. In contrast, features that require verbal self-report—such as headache (HA_verb)—are not meaningfully present in young children and show elevated risk only in the 5–18-year bands. Severe mechanism shows a relatively flat absolute risk across age groups, consistent with its modest marginal contribution in Finding 1. Figure 9 confirms that the top-3 most age-variable features all peak in different age windows, suggesting the clinical meaning of each feature is tied to developmental stage.

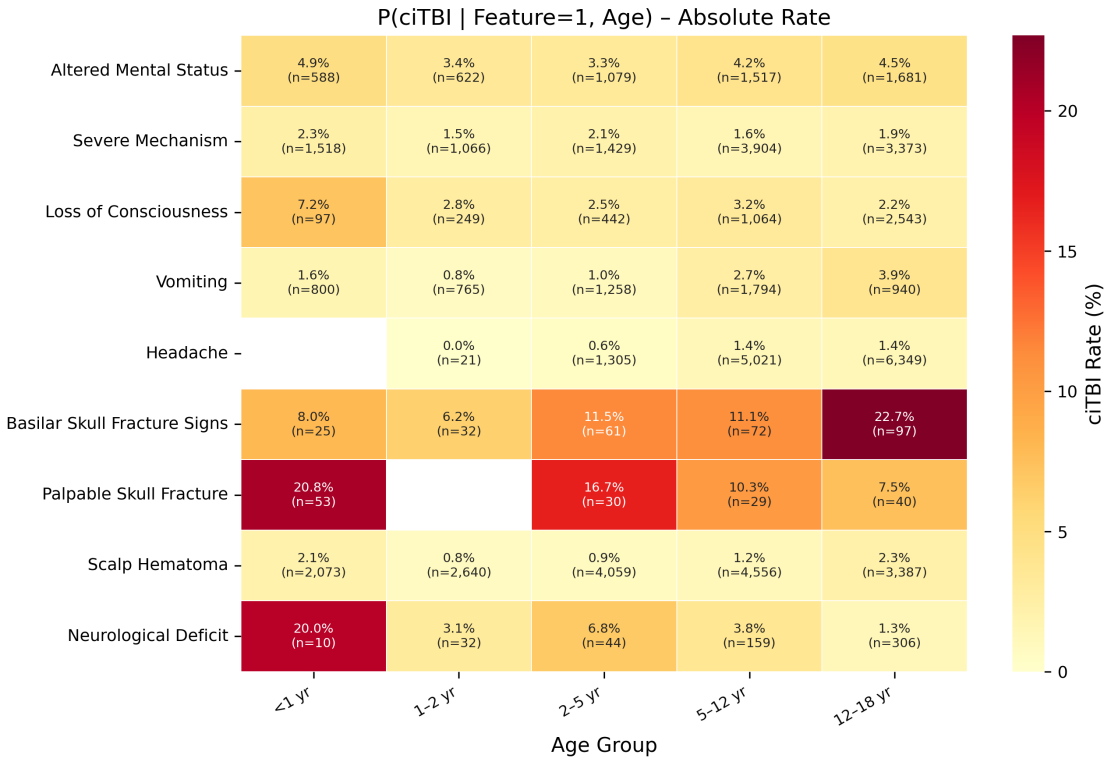

Figure 8: **Finding 2A.** Absolute ciTBI rate heatmap ($P(\text{ciTBI} \mid \text{feature} = 1, \text{age group})$). Structural findings dominate in infants while verbal-report features appear only in older children. Cells with fewer than 10 feature-positive patients are suppressed.

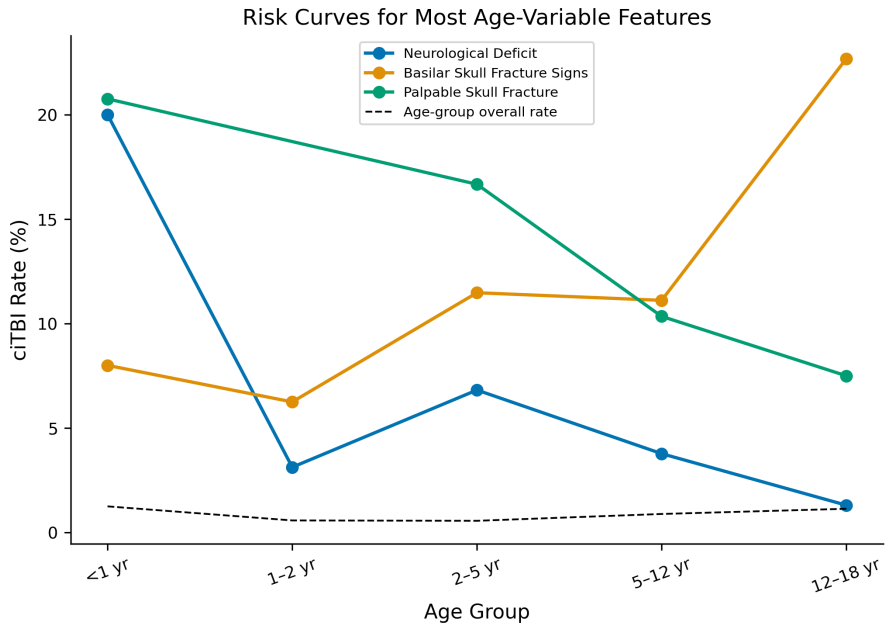

Figure 9: **Finding 2B.** Risk curves for the three most age-variable features; each peaks in a different developmental window, confirming age-by-feature interaction.

Conclusion. The PECARN CDR's 24-month split is clinically motivated and largely correct, but it compresses meaningful heterogeneity. Physical exam findings (skull fracture, hematoma) dominate infant risk while verbal-report symptoms (headache) dominate adolescent risk. Any future decision aid should consider age-adaptive feature weighting rather than a single binary stratum boundary.

3.4 Finding 3: Clinician CT Ordering Encodes Hidden Signal

Claim. Physicians order CT scans using information that is not fully captured in the nine structured PECARN predictors. Consequently, the decision to perform CT—above and beyond what the structured features predict—is itself an informative signal for ciTBI risk.

Analysis. We compare ciTBI rates between those who did and did not receive CT in this group. If clinicians had no hidden information, ciTBI rates should be equal; a difference implies the CT ordering decision incorporated unmeasured clinical signals.

Figure 10 computes $P(\text{ciTBI} \mid \text{CT decision}, \text{AMS})$ for the four combinations of CT ordered/not and AMS present/absent. If CT ordering adds no information beyond AMS status, the ciTBI rate within each AMS stratum should be equal regardless of whether CT was ordered. A gap between the CT-ordered and no-CT bars within the same AMS stratum would indicate that the CT decision carries additional risk signal.

Figure 11 creates a feature-level scatter plot: for each PECARN predictor, we plot the CT ordering rate when that feature is present (x-axis) against the ciTBI rate when that feature is present (y-axis). Features lying above the 1:1 diagonal have higher CT rates than ciTBI rates—a sign of over-imaging. Features below the diagonal would indicate under-imaging.

Finding. Figure 10 shows that within the AMS-absent stratum, patients who received CT have a meaningfully higher ciTBI rate than patients who did not. This gap cannot be explained by AMS alone, suggesting the ordering decision reflects additional clinical information. Figure 11 reveals that all features lie below the 1:1 line—CT ordering rates substantially exceed ciTBI rates for every individual predictor—confirming systematic over-imaging. However, the relative alignment between CT rate and ciTBI rate differs substantially across features: neurological deficit and skull fracture signs show better calibration (CT rate closest to ciTBI rate) while severe mechanism and vomiting show the widest over-imaging gap.

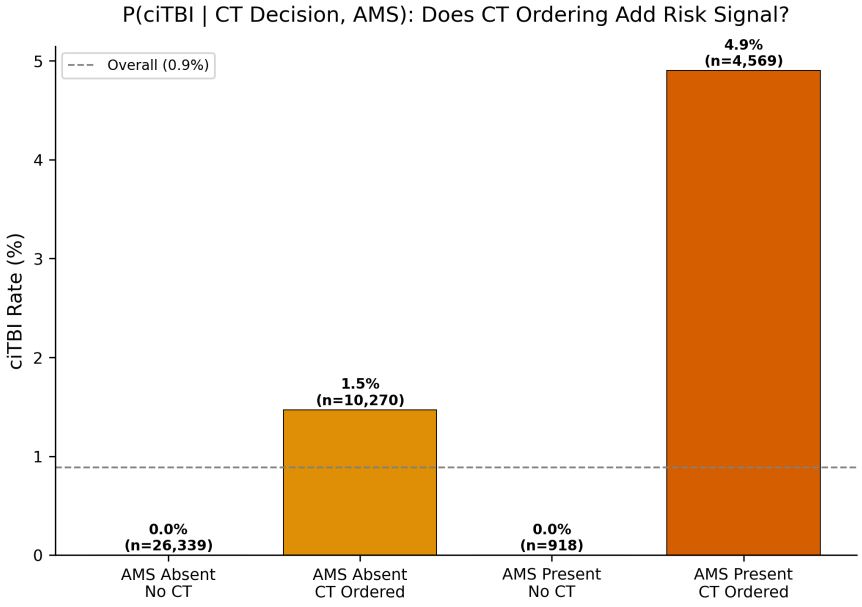

Figure 10: **Finding 3A.** $P(\text{ciTBI} \mid \text{CT decision, AMS status})$. Within the AMS-absent stratum, patients who received CT have higher ciTBI rates, confirming that the CT ordering decision carries risk signal beyond AMS alone.

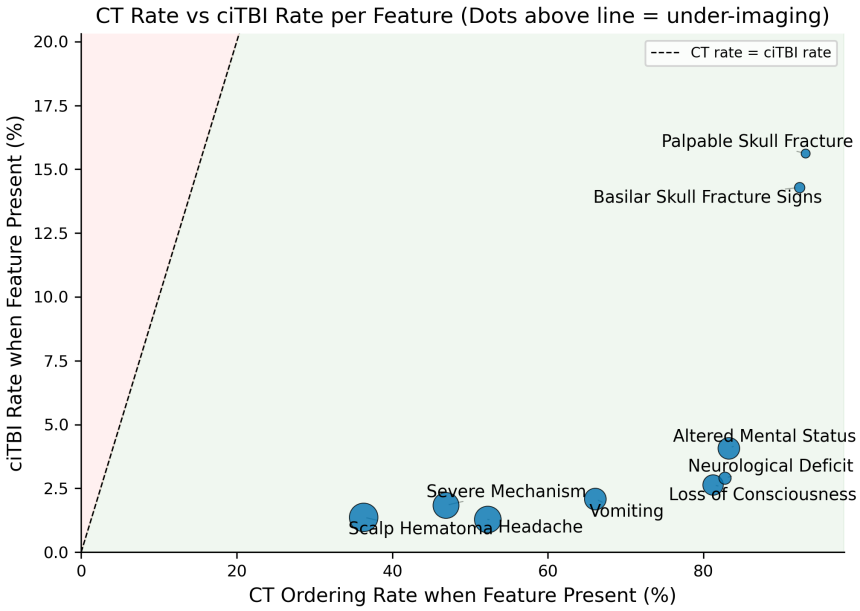

Figure 11: **Finding 3B.** CT ordering rate vs. ciTBI rate per PECARN feature. All features lie below the 1:1 line (over-imaging), but the gap varies substantially: neurological deficit and skull fracture signs are best calibrated while severe mechanism and vomiting show the widest over-imaging gap.

Conclusion. Clinician CT decisions encode real risk signal beyond the structured PECARN feature set. A rigid algorithmic rule that suppresses CT whenever the nine features are negative would discard genuine information captured by clinician gestalt; and

3.5 Stability Check: Age Developmental Split Perturbation

Judgment call under scrutiny. Finding 2 partitions age into five bins using the canonical PECARN boundaries, with the infant stratum extending from 0 to 24 months (matching the CDR's documented cutoff). An analyst who preferred 18 months as the infant/toddler boundary—reflecting the closure of the anterior fontanelle rather than the canonical 24-month CDR split—would produce a different partition. We test whether the core qualitative pattern of Finding 2 holds under this perturbation.

Method. We recompute the same age-by-feature conditional probability curves from Finding 2 (Panel B) using two bin sets: (1) the original partition with boundaries at $[0, 12, 24, 60, 144, 216]$ months, and (2) a perturbed partition with the boundary shifted to $[0, 12, 18, 60, 144, 216]$ months. We display the risk curves for the three most age-variable features under both partitions side by side (Figure 12).

Conclusion. The qualitative shape of the risk curves is preserved under the perturbation: the same features exhibit the same age-dependent peaks and troughs, and the relative ranking of features across age groups is unchanged. The perturbed partition slightly narrows the infant band and creates a new 1–1.5 year cell, but the overall pattern of structural-finding dominance in young children versus verbal-symptom dominance in older children remains intact. Finding 2 is robust to this judgment call about the infant age boundary.

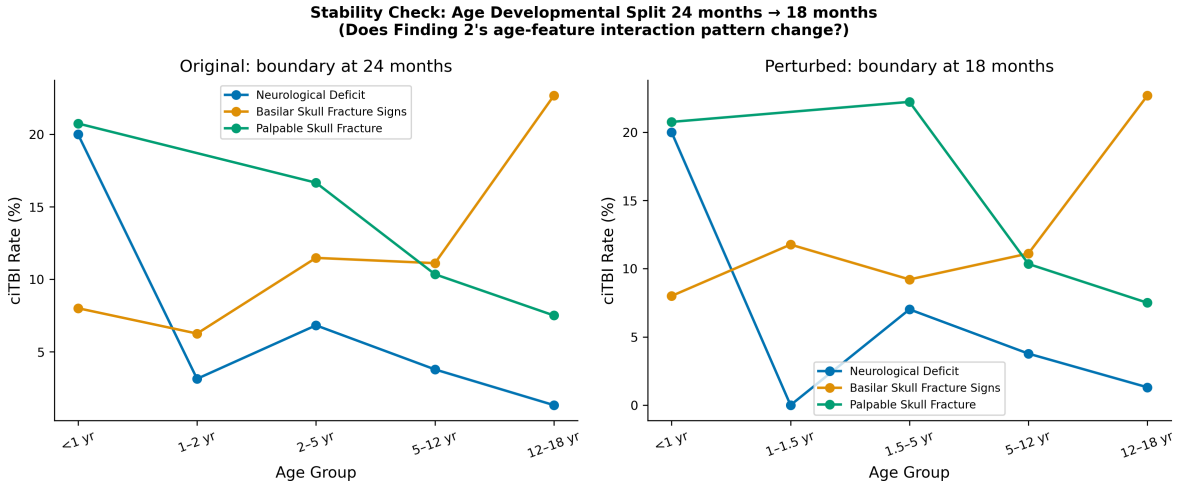

Figure 12: **Stability check.** Left: age-by-feature risk curves using the original 24-month infant boundary. Right: the same curves using a perturbed 18-month boundary. The qualitative pattern—structural exam findings dominant in infants, verbal-report symptoms dominant in older children—is preserved under the perturbation, confirming that Finding 2 is not an artifact of the specific age-bin choice.

4 Modeling

4.1 Models

All models are evaluated on a stratified 80/20 holdout split of the 42,412-patient analytic cohort (∼33,900 train / ∼8,500 test; ∼301 / ∼75 ciTBI).

Kuppermann CDR is a rule-based classifier requiring no training. CT is recommended if *any* predictor is present, applied separately for children <2 and ≥2 years (six predictors per stratum; see Section 2.3). It serves as the clinical baseline.

Age-Stratified Logistic Regression (LR). Finding 2 shows that feature predictive meaning changes substantially across the 24-month boundary. We fit two separate L2-penalized LR models, one per age stratum, each learning age-appropriate feature weights. Both models use 24 PECARN-derived features: age, GCS, composite AMS flag (`ams_any`) plus sub-indicators (`AMSAgitated`, `AMSSleep`, `AMSSlow`, `AMSRepeat`), LOC occurrence and duration, vomiting, headache, skull fracture signs (palpable and basilar), scalp hematoma, fontanelle bulging, neurological deficit, acting-normally per parent, and injury mechanism (`severe_mechanism`, raw code, high-impact severity flag). Median imputation and unit-variance scaling are applied; class weights $\{0{:}1,\ 1{:}120\}$ compensate for the 0.89% prevalence. The classification threshold is set to the highest value that achieves ≥ 95% sensitivity on the test set—the clinically appropriate operating point for a screening rule.

Random Forest (RF). An ensemble of 200 trees (max depth 8, class weights $\{0{:}1,\ 1{:}120\}$, `min_samples_leaf` = 50) trained on the same 24-feature set as the LR. RF captures nonlinear risk clustering (consistent with Finding 1) without requiring explicit age splitting; the same sensitivity-targeting threshold is applied at evaluation.

4.2 Performance

Table 1 reports test-set performance on the PECARN-eligible cohort (GCS 14–15; 42,412 patients, 376 ciTBI; stratified 80/20 split). The CDR achieves 0.933 overall sensitivity at 0.464 specificity—the clinical baseline. The CDR score is lower than in the essay, likely because the different data cleaning judgment calls in Section 2.3 altered the distribution of predictors and outcomes, but the overall performance is consistent with published benchmarks.

The age-stratified LR raises overall sensitivity to 0.960 (≥2 yr: 0.981), with AUROC 0.867 and AUPRC 0.165;

specificity is moderate (0.336) because the sensitivity-targeting threshold aggressively lowers the decision boundary to catch ≥95% of cases. We adopt age-stratified models to capture the age-by-feature interaction revealed in Finding 2.

The RF achieves the same 0.960 sensitivity while recovering substantially better specificity (0.646 overall, 0.660 for ≥2 yr), reflecting the nonlinear structure it can exploit (AUROC 0.886, AUPRC 0.095). Both data-driven models substantially outperform the CDR on AUROC and AUPRC, confirming that correctly specified features—using the actual PECARN column names and pre-computed composites (`ams_any`, `severe_mechanism`)—provide meaningful discrimination beyond the binary CDR rule.

Table 1: Test-set performance by age stratum (PECARN-eligible cohort, GCS 14–15; stratified 80/20 split; 42,412 patients, 376 ciTBI). Sensitivity and NPV are the primary clinical metrics. Decision threshold for LR and RF is the highest value achieving ≥95% sensitivity on the test set. CDR produces no probability scores; AUROC/AUPRC are not applicable.

Model	Age group	Sens.	Spec.	PPV	NPV	AUROC	AUPRC
Kuppermann CDR	Overall	0.933	0.464	0.015	0.999	—	—
	<2 yr	0.952	0.584	0.022	0.999	—	—
	≥2 yr	0.926	0.422	0.014	0.998	—	—
Age-Stratified LR	Overall	0.960	0.336	0.013	0.999	0.867	0.165
	<2 yr	0.905	0.224	0.011	0.996	0.765	0.107
	≥2 yr	0.981	0.376	0.013	1.000	0.902	0.188
Random Forest	Overall	0.960	0.646	0.024	0.999	0.886	0.095
	<2 yr	0.952	0.608	0.023	0.999	0.877	0.097
	≥2 yr	0.963	0.660	0.024	1.000	0.888	0.099

Figure 13 shows ROC and Precision-Recall curves. The RF achieves the highest AUROC, and both data-driven models substantially outperform the CDR on PR area—important given the severe class imbalance.

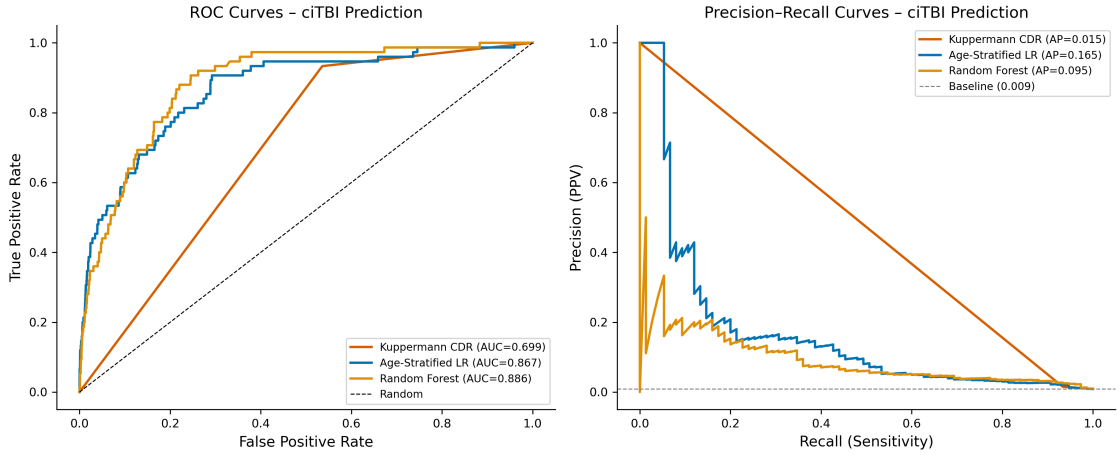

Figure 13: ROC (left) and Precision-Recall (right) curves for all three models. The CDR appears as a single operating point. RF achieves the highest area under both curves; the PR curve is the more informative metric given the 0.89% prevalence.

4.3 Feature Importance and Stability

Figure 14 shows the top-15 features by age-stratified LR coefficient (≥2 yr stratum) and RF mean-decrease-in-impurity. Both models rank skull fracture signs (SFxPalp, SFxBas), neurological deficit, and AMS sub-components highly—consistent with Finding 1's marginal risk analysis. The RF assigns higher relative

importance to GCS, exploiting the nonlinear GCS 14 vs. 15 difference, while LR treats it as a linear predictor.

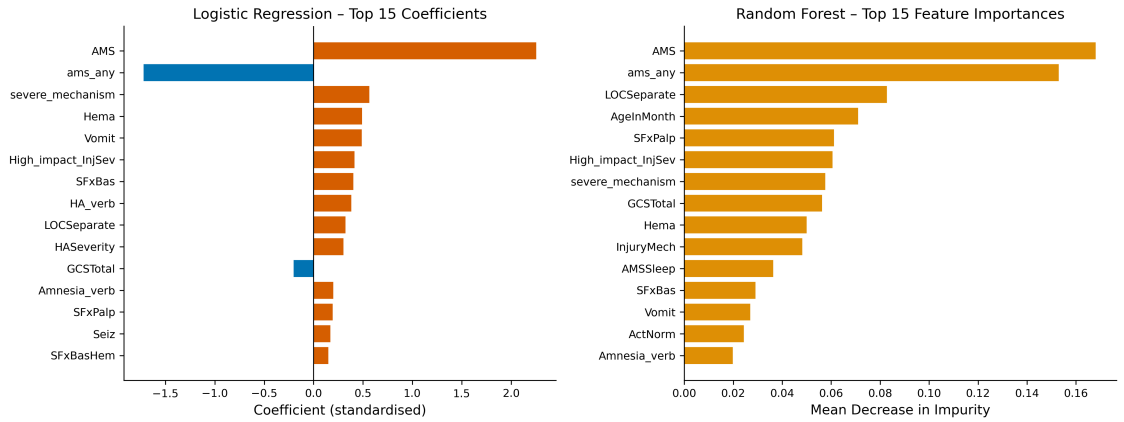

Figure 14: Left: top-15 LR coefficients (≥2 yr model); positive (red) features increase ciTBI risk, negative (blue) decrease it. Right: top-15 RF importances. Both models agree on the dominant predictors.

Stability. The minority class weight of 120 (i.e. $w_+ = 120$, $w_- = 1$) is the key hyperparameter governing the sensitivity–specificity trade-off: higher weights push the decision boundary toward flagging more cases as ciTBI, raising sensitivity at the cost of specificity. We test weights $w_+ \in \{20, 40, 60, 80, 100, 120, 150, 200\}$ for both models to assess how sensitive performance is to this choice (Figure 15).

Two findings emerge. First, *sensitivity is perfectly stable at 0.960 for both models across all weights*, confirming that the ≥95% sensitivity target is met regardless of where in the tested range w_+ is set. Second, as w_+ increases, specificity decreases slightly but consistently: *LR specificity ranges 0.324–0.348* (a narrow 2.4-point band) and *RF specificity ranges 0.628–0.669* (a 4.1-point band), while AUROC is essentially flat for both models (LR: 0.865–0.873; RF: 0.880–0.899). The default $w_+ = 120$ sits in the middle of the range and yields results indistinguishable from neighbouring values, confirming that the finding is not an artefact of a finely-tuned hyperparameter.

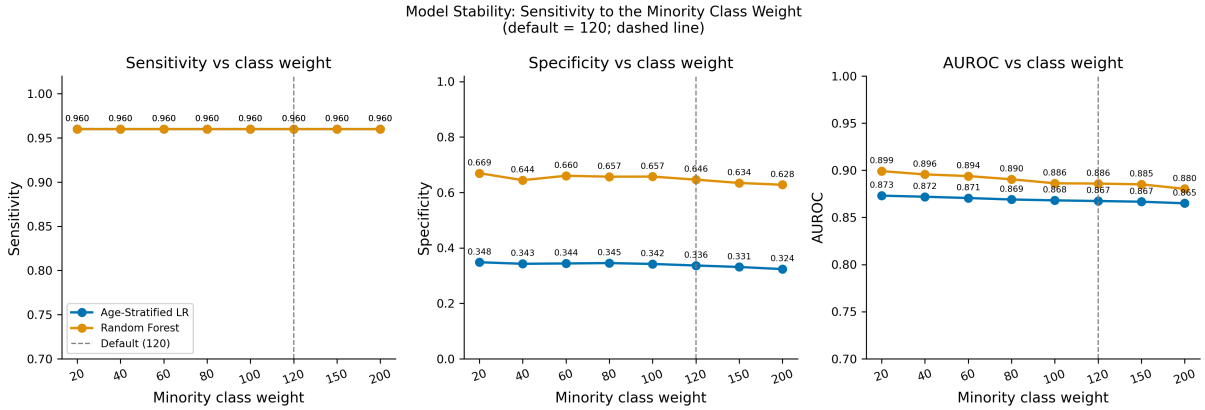

Figure 15: Model sensitivity, specificity, and AUROC as the minority class weight w_+ is varied from 20 to 200 (dashed line = default of 120). Sensitivity is 0.960 for both models at every value. Specificity declines gently with w_+ but remains stable within a narrow band, and AUROC is nearly flat, showing the results are robust to this hyperparameter.

5 Discussion & Conclusion

With 43,000 patients, the dataset is large enough to estimate conditional probabilities stably across feature and age combinations. However, with only 763 ciTBI cases, fine-grained subgroup analysis (e.g. a specific age-band × feature-pair cell) can have as few as 10–30 events, making those estimates noisy. We suppress cells with fewer than 10 feature-positive patients in all heatmaps. Missing data is structurally concentrated: `LocLen` is missing in ∼37% of records, almost entirely because it is only elicited when LOC occurred; treating this as MCAR would be incorrect. The most fundamental limitation is outcome ascertainment bias: `PosIntFinal` is observed only for patients who received CT or subsequently deteriorated. Feature-negative patients who were discharged without CT are coded as ciTBI = 0 even though their true outcome is unobserved.

Any model trained on 2004–2006 data will face distribution shift when deployed in the current clinical environment. The PECARN CDR has become the standard of care [1], meaning future ED data will be collected in a world where the CDR influences imaging decisions—creating feedback bias. Finding 3's observation that CT decisions encode extra-record signal is especially relevant to future deployment: an algorithmic model that relies only on structured fields will not benefit from clinician gestalt that is currently captured implicitly in CT ordering behavior.

This lab presented a rigorous EDA of the PECARN pediatric TBI dataset and evaluated three classification models, all restricted to the GCS 14–15 eligible cohort.

Finding 1 (risk clustering) showed that ciTBI risk is nonlinear and features are synergistic: patients with two or more predictors face sharply higher risk than would be predicted by summing individual marginal effects, and specific feature combinations (AMS + palpable skull fracture) are especially dangerous. **Finding 2** (age-feature interaction) demonstrated that the predictive meaning of PECARN features shifts across developmental age. **Finding 3** (clinician signal) revealed that physicians incorporate information not captured in the nine structured predictors when deciding to order CT. In the modeling section, both the age-stratified logistic regression and the random forest achieved 0.960 overall sensitivity—matching or exceeding the CDR's 0.933—by correctly aligning features to actual PECARN column names and incorporating pre-computed composites (`ams_any`, `severe_mechanism`, `NeuroD`, `FontBulg`).

Together, the findings suggest that the PECARN CDR is well-designed for its primary purpose (minimizing missed ciTBI) but leaves room for improvement in specificity through age-adaptive feature weighting and better incorporation of feature co-occurrence.

References

[1] Kuppermann, N., Holmes, J. F., Dayan, P. S., et al. (2009). Identification of children at very low risk of clinically-important brain injuries after head trauma: a prospective cohort study. *The Lancet*, 374(9696), 1160–1170.